

TUMELO MAHLOGONOLO EMERALD

MOHALE

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PROGRAMMING ASSIGNMENT

RaceDay - API Endpoint Plan

Legend

- **Role Required:** None (public), Any (any logged-in user), Organizer, or Participant
- All authenticated routes expect Authorization: Bearer <token>

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as an Organizer or Participant.	None	{ fullName, email, password, role, phoneNumber }	201 Created - user id and email; 400 Bad Request - validation error; 409 Conflict - email already in use
POST	/api/auth/login	Authenticates a user and returns a JWT access token and refresh token.	None	{ email, password }	200 OK - { accessToken, refreshToken, role }; 401 Unauthorized - invalid credentials
POST	/api/auth/refresh	Issues a new access token using a valid refresh token.	None	{ refreshToken }	200 OK - new { accessToken }; 401 Unauthorized - expired/invalid token
POST	/api/auth/logout	Revokes the supplied	Any	{ refreshToken }	204 No Content; 401 Unauthorized

		refresh token.			
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2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the logged-in user's profile.	Any	None	200 OK - user object; 401 Unauthorized
PUT	/api/users/me	Updates the logged-in user's profile details.	Any	{ fullName, phoneNumber }	200 OK - updated user object; 400 Bad Request; 401 Unauthorized
PUT	/api/users/me/password	Changes the logged-in user's password.	Any	{ currentPassword, newPassword }	200 OK; 400 Bad Request - wrong current password; 401 Unauthorized

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all published events (supports filtering by type, date, province).	None	None	200 OK - array of events
GET	/api/events/{id}	Returns full details of a single event,	None	None	200 OK - event object;

		including its venue and categories			404 Not Found
POST	/api/events	Creates a new event owned by the logged-in Organizer.	Organizer	{ name, description, eventDate, eventType, venue: { addressLine, city, province, latitude, longitude } }	201 Created - event object; 400 Bad Request; 401/403
PUT	/api/events/{id}	Updates an event owned by the logged-in Organizer.	Organizer	{ name, description, eventDate, eventType, status }	200 OK - updated event; 403 Forbidden - not the owner; 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organizer.	Organizer	None	204 No Content; 403 Forbidden; 404 Not Found
GET	/api/events/{id}/weather	Returns live weather forecast for the event's venue location/date.	None	None	200 OK - weather summary; 404 Not Found
GET	/api/events/mine	Lists all events created by the logged-in Organizer.	Organizer	None	200 OK - array of events

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories for a specific event.	None	None	200 OK - array of categories ; 404 Not Found
POST	/api/events/{eventId}/categories	Adds a new category to an event owned by the logged-in Organizer.	Organizer	{ name, distanceKm, maxParticipants, entryFee }	201 Created - category object; 403 Forbidden ; 404 Not Found
PUT	/api/categories/{id}	Updates a category belonging to the Organizer's event.	Organizer	{ name, distanceKm, maxParticipants, entryFee }	200 OK - updated category; 403 Forbidden ; 404 Not Found
DELETE	/api/categories/{id}	Deletes a category (only if it has no enrolments).	Organizer	None	204 No Content; 403 Forbidden ; 409 Conflict - has enrolments

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/enrol	Enrols the logged-in Participant	Participant	None	201 Created - enrolment with bib

		into a category			number; 400 Bad Request - category full; 401 Unauthorized; 409 Conflict - already enrolled
GET	/api/enrolments/mine	Lists all enrolments (past and upcoming) for the logged-in Participant.	Participant	None	200 OK - array of enrolments
DELETE	/api/enrolments/{id}	Cancels the logged-in Participant's own enrolment.	Participant	None	204 No Content; 403 Forbidden - not the owner; 404 Not Found
GET	/api/events/{eventId}/enrolments	Lists all enrolments for an event, for the owning Organizer.	Organizer	None	200 OK - array of enrolments; 403 Forbidden

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures a race result for a participant'	Organizer	{ finishTime , overallPosition, categoryPo	201 Created - result object;

		s enrolment.		sition, raceStatus }	403 Forbidden; 404 Not Found; 409 Conflict - result already exists
PUT	/api/results/{id}	Updates/cor rects a previously captured result.	Organi zer	{ finishTime , overallPosit ion, categoryPo sition, raceStatus }	200 OK - update d result; 403 Forbidden; 404 Not Found
GET	/api/events/{eventId}/res ults	Lists all results for an event (public leaderboar d)	None	None	200 OK - array of results
GET	/api/results/mine	Returns the logged-in Participant' s full personal result history.	Partici pant	None	200 OK - array of results